

Paired comparisons with quiet surface drones show evidence of fish behavioral response to motorized vessels during acoustic surveys in Lake Superior

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Abstract

Acoustic surveys are important for fish stock assessments, but fish responses to survey vessels can bias acoustic estimates. We leveraged quiet uncrewed surface vessels (USVs) to characterize potential bias in acoustic surveys. Five conventional motorized ships overtook USVs from astern over 2 km transects at night in Lake Superior in 2022. We examined the difference in acoustic backscatter, average target depth, and average target strength (TS) between USV and motorized vessels. Although sound level measurements from the motorized vessels sometimes exceeded recommendations for scientific vessels, we did not detect differences in acoustic measures among survey vessels. However, the USVs recorded 2 dB higher acoustic backscatter and TS than motorized vessels, leading to ~15% higher fish densities with drones when using in situ TS and echo integration. Differences in fish density would increase to 30%–60% if a standard TS value was applied. Target depth did not differ between USVs and motorized ships. These results are consistent with a change in orientation but not depth of insonified fish and limited horizontal avoidance of motorized survey vessels.

Key words: acoustic survey, vessel avoidance, autonomous vehicle, drone

Introduction

Acoustic surveys are important tools for assessing fish populations and informing fisheries management decisions as they allow fisheries-independent data to be collected over large spatial scales (Simmonds and MacLennan 2005; Rudstam et al. 2012). The accuracy of acoustic surveys depends on their ability to collect unbiased data, and therefore an understanding of how fish behavioral responses to survey vessels influence these abundance estimates but remains a knowledge gap (Olsen 1990; De Robertis and Handegard 2013). Prior work has documented that fish can detect and respond to survey vessels, potentially biasing population estimates and affecting the accuracy of acoustic assessments

(Misund 1990; Vabø et al. 2002; De Robertis and Handegard 2013).

Fish responses to survey ships are assumed to be primarily the result of vessel noise generated by combustion engines and propellers that vary among vessels (Mitson 1995; De Robertis et al. 2008). As a result, ship avoidance may vary among survey ships resulting in potential biases when combining data from multiple ships in a single survey. However, the effect of ship noise on acoustic surveys is uncertain and measuring these effects in situ has proven difficult (Fernandes et al. 2000; De Robertis and Wilson 2010; DuFour et al. 2021; Elliott et al. 2021). In the Laurentian Great Lakes, acoustic surveys are conducted on each lake by multiple mo-

torized vessels due to the large size of these lakes and operational limitations of the vessels used (Parker-Stetter et al. 2009; Yule et al. 2012; Warner et al. 2022). Therefore, differential vessel avoidance by fishes could lead to assessment bias, making it important to understand whether fish avoidance varies between vessels and to what degree data from different vessels are comparable.

The advent of novel drone-based technologies provides opportunity to assess the effects of ship noise on fisheries acoustics estimates by comparing motorized vessels against silent platforms. Autonomous, and effectively silent, high endurance uncrewed surface vessels (USVs) fitted with acoustic transducers are commercially available and their potential for fisheries surveys is significant (De Robertis et al. 2021; Totland and Johnsen 2022; Xu et al. 2023). Uncrewed surface vehicles are often smaller than traditional motorized survey vessels and can be designed to minimize cues that may alter fish behavior and distributions in their vicinities. Wind propelled USVs are especially quiet (Evans et al. 2023) and unlikely to elicit fish responses. These effectively silent USVs provide an “unimpacted” acoustic signal benchmark and enable novel opportunities to assess potential bias in conventional motorized survey vessels. Here, we leveraged a USV (“Saildrone”, Saildrone, Inc., Alameda, CA, USA) propelled by sail and powered by solar panels to test for fish responses to motorized survey vessels in Lake Superior, the largest of the Laurentian Great Lakes.

To test whether fish avoid the combustion powered ships used in Lake Superior, we deployed two identical quiet USVs and compared their acoustic data against five traditional motorized vessels in western Lake Superior in summer 2022. A similar study was conducted in 2021 in lakes Huron and Michigan (Evans et al. 2023), but unlike all the other Great Lakes, the native *Coregonus* species flock (*Coregonus* spp.) remains largely intact in Lake Superior (Eshenroder et al. 2016). The pelagic species cisco (*Coregonus artedii*) is of key interest to Lake Superior management agencies because this population supports a commercial fishery with recent annual lake-wide yields of 1000 metric tons (Goldsworthy and Yule 2021). Moreover, acoustic surveys of cisco are used to set harvest quotas in multiple jurisdictions (Goldsworthy and Yule 2021) so research to understand how native pelagic prey fish in Lake Superior respond to approaching research vessels represents an important undertaking because different species are known to behave differently to ship noise. For instance, Alaskan pollock (*Gadus chalcogrammus*) and Atlantic herring (*Clupea harengus*) showed opposite responses to noise reduced vessels (De Robertis and Handegard 2013). The USVs we used are small vessels (~7 m) that rely on wind for propulsion and solar power to operate onboard equipment. Noise generated by these USVs was not detectable over ambient noise in a similar survey in lakes Michigan and Huron (Evans et al. 2023). All USVs and vessels were equipped with 120 or 123 kHz split beam echosounders. Our objectives were to (1) characterize the sound emanating from the motorized survey vessels and USVs and compare them to noise standards proposed for acoustic survey vessels by International Council for the Exploration of the Sea (ICES; Mitson 1995); (2) evaluate whether fish exhibit a behavioral response (either swimming away from

the ship or diving) in front of approaching motorized vessels by quantifying changes in acoustic measurements from the quiet USV as motorized survey vessels approached; and (3) assess whether fish abundance estimates differ between the USV and the motorized ships. We hypothesized that as vessels approached the USVs, fish would show a behavioral response that would lead to a decrease in their overall abundance (measured by nautical area scattering strength; S_A), average target depth (TD), and average target strength (TS), and these decreases would be related to the noise level of the overtaking survey vessel. We also expected the USVs to measure higher S_A , shallower average TD, higher TS, and higher fish densities than motorized vessels when comparing fishes distributed in the same 2 km transects sampled on the same night. Lastly, we hypothesized we would see the strongest responses in the upper waters as vessel noise should be louder at these depths and prior investigations into fish behavior in Lake Superior have indicated the strongest avoidance in water shallower than 40 m (Grow et al. 2020).

Materials and methods

Study system

Lake Superior is the largest and deepest of the Laurentian Great Lakes and has a mixed assemblage of pelagic fishes that are distributed throughout the water column (Vinson et al. 2023). The upper water column (~50 m) is utilized by adult rainbow smelt (Osmeridae: *Osmerus mordax*), adult cisco (*Coregonus artedii*), and other juvenile coregonines (*Coregonus* spp.; Yule et al. 2013). Deeper waters harbor both adult bloater (*Coregonus hoyi*) and kiyi (*Coregonus kiyi*; Rosinski et al. 2020). Finally, large predatory salmonids, mainly lake trout (*Salvelinus namaycush*), are distributed throughout the water column, although these fish are less abundant than smelt and coregonines.

Study vessels

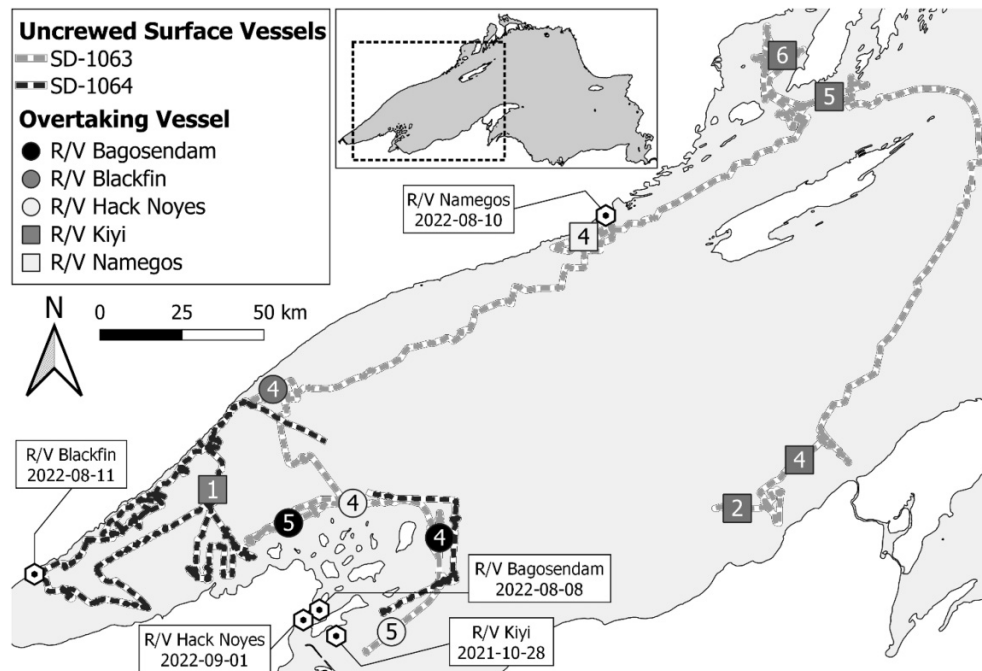
The two USVs used in this study had identical configurations. Each drone was 7 m in length and had a transducer mounted on a gimbal on the bottom of the keel (2 m deep) that compensated for vessel roll. The drones were propelled by wind, which was captured by a 5 m wing, and relied on solar panels to supply battery power for equipment onboard. Each drone was equipped with a 120 kHz split-beam transducer (Simrad EK80) with a beam angle of 7°. (Table 1). Pulses were a continuous wave (CW) with a pulse duration of 0.256 ms and a ping rate of 1.31 pings/s (Table 1).

Five motorized survey vessels were used in this study: R/V *Bagosendam*, R/V *Blackfin*, R/V *Hack Noyes*, R/V *Kiyi*, and R/V *Namegos*. All vessels were powered by diesel engines. Great Lakes fish taxa exhibit changes in behavior during daytime hours that can affect acoustic measurements, including diel migrations into acoustic dead zones and schooling during the day. Thus, current acoustic fisheries surveys in the Great Lakes occur exclusively at night. Accordingly, comparisons of USVs and motorized vessels in this study were also conducted during nighttime hours. Overtakes of the USVs occurred on

Table 1. Vessel and echosounder characteristics from acoustic survey testing on Lake Superior in 2022.

	R/V <i>Bagosendam</i>	R/V <i>Blackfin</i>	R/V <i>Hack Noyes</i>	R/V <i>Kiyi</i>	R/V <i>Namegos</i>	SD-1063/SD-1064
Type	Motorized vessel	Motorized vessel	Motorized vessel	Motorized vessel	Motorized vessel	Uncrewed surface vessel
Length (m)	11.3	9.4	17	32.6	9.1	7.0
Beam (m)	3.4	2.6	4.4	8.2	3.6	1.0
Draft (m)	0.4	0.9	1.8	2.9	0.6	2.0
Number of engines	2	2	1	1	2	0
Number of blades on prop	3	3	3	4	3	0
Shaft revolutions per minute during survey	800–1100	600	700	650	850	NA
Mean survey speed (m/s) (mean $\pm$ standard deviation)	2.5 ($\pm$ 0.1)	2.5 ($\pm$ 0.8)	2.8 ($\pm$ 0.1)	2.8 ($\pm$ 0.2)	2.1 ($\pm$ 0.03)	0.7 ($\pm$ 0.3)/0.3 ($\pm$ 0.1)
Hydraulic status during acoustic survey	Off	Off	Off	On	Off	NA
Mean 200 m approach noise level dB re 1 μ Pa (variation between runs—dB)	121.0 (Range: 19.5)	84.1 (Range: 14.0)	118.2 (Range: 10.6)	97.2 (Range: 4.0)	120.5 (Range: 13.0)	NA
Mean closest pass noise level dB re 1 μ Pa (variation between runs—dB)	131.6 (Range: 19.3)	110.0 (Range: 31.2)	120.8 (Range: 11.9)	112.7 (Range: 20.5)	131.2 (Range: 9.5)	NA
Acoustic deck unit	BioSonics DTX	BioSonics DTX	BioSonics DTX	BioSonics DTX	BioSonics DTX	Simrad EK80
Transducer mounting	Biofin tow fish	Pole mounted	Pole mounted	Moon pool	Pole mounted	Keel
Transducer depth (m)	0.84	1.0	0.84	2.3	1.0	2.0
Frequency (kHz)	120	123	120	120	120	120
Source level (dB re 1 μ Pa)	218.5	221.2	219.1	218.9	221.1	209
Pulse duration (ms)	0.400	0.400	0.400	0.400	0.400	0.256
Ping rate (pings/s)	3	1	3	2, 1, 0.65	0.8	0.76
Background S_V noise dB re 1 m^{-1} @ 1 m	–134	–129	–115	–123 to –132	–135	–134 to –135
Calibration sphere material	Tungsten carbide	Tungsten carbide	Tungsten carbide	Tungsten carbide	Tungsten carbide	Tungsten carbide
Calibration sphere size (mm)	33.2	33.2	33.2	33.2	33.2	38.1
Calibration date(s)	4 August 2022, 12 August 2022	19 August 2022, 6 September 2022	6 July 2023	28 August 2022, 8 September 2022, 19 September 2022	26 August 2022	8 August 2022 (SD-1063), 6 August 2022 (SD-1064), 28 September 2022(both)

Fig. 1. Map of uncrewed surface vessel (SD-1063, SD-1064) tracks and overtakes by powered vessels in Lake Superior during the summer of 2022 with the number of overtakes that occurred at these points in the center of each symbol. In addition, the locations where powered vessels were sound profiled are marked by hexagons; the profiled vessel and date when the profile occurred are noted adjacent to each site. Lake Superior contour was downloaded from ScienceBase. All projections were in WGS84, datum: EPSG 4326.



separate nights and in different regions of Lake Superior (Fig. 1). Ships approached a drone from astern while operating under normal acoustic survey settings (e.g., speed). Survey vessels utilized either a BioSonics 120 or 123 kHz split-beam transducer (Table 1), which were different from the transducers used by the USVs. We considered data collected by the Simrad and BioSonics transducers to be comparable as all units were calibrated and because prior studies with both types of echosounders show comparable results (Wanzenböck et al. 2003; Draščík et al. 2017). All ship transducers used a 0.400 ms CW pulse, which is the standard pulse duration used in the Great Lakes (Parker-Stetter et al. 2009); pulse rates were variable (Table 1). The difference in pulse duration between USV and motorized vessel is considered in the discussion. Transmitted power was ≤ 250 W for all transducers, and therefore nonlinear loss to higher harmonic frequencies was negligible.

The USV transducers were calibrated at the start and end of each deployment with a 38.1 mm tungsten carbide sphere. Vessel transducers were calibrated with a 33.2 mm tungsten carbide sphere once for the R/V Hack Noyes and R/V Namegos and twice for all other vessels (Table S1). Returns were analyzed with the calibration assistant available in Echoview v13.1 to obtain beam angles, beam offsets, and S_A offsets and transducer gain for Simrad units or S_v and TS offsets for BioSonics units. For the USVs, R/V Bagosendam, R/V Blackfin, and R/V Kiyi we used the first calibration conducted and confirmed this calibration was stable by comparing it to the second calibration (Table S1). The R/V Hack Noyes was calibrated

in the summer of 2023 because earlier calibration attempts in 2022 yielded few (<10) on-axis targets.

Vessel acoustic profiling

The five motorized survey vessels were profiled in Lake Superior for radiated noise using the same procedure and data processing outlined in Evans et al. (2023) in test locations around the Apostle Islands; offshore of Ashland, Wisconsin; near Grand Portage, Minnesota; or offshore of Duluth, Minnesota (Fig. 1). Briefly, powered vessels ran a 2 km straight-line course through a hydrophone array that measured the radiated noise generated by the vessels. The hydrophone array consisted of three vertically oriented hydrophones moored 2 m from the bottom and spaced at 1 m depths in 30 m of water (Evans et al. 2023).

Sound speed profiles were recorded at each measurement location. The mean sound speeds were similar for all survey vessels ($1424\text{--}1468\text{ m}\cdot\text{s}^{-1}$) and variations in sound speed from surface to bottom were small (0.32%–2.65%). The sound speed profiles all monotonically decreased with depth indicating a slight downward refracting condition. Because of the relatively small change in sound speed with depth, combined with the close measurement range, low frequency of the vessel noise, and variability in background at test sites, corrections for the acoustic channel were not done and received levels are reported here for comparison. Each measurement was averaged for 10 s during which the vessels were moving, which helped to mitigate, but not eliminate, the acoustic

channel effects through averaging. The acoustic signatures of the motorized vessels were compared to the underwater radiated noise specification proposed by [Mitson \(1995\)](#).

The USV noise was not measurable above the surface wind- and wave-generated background noise when underway ([Evans et al. 2023](#)). Therefore, radiated noise measurements were recorded dockside (i.e., the USV was stationary) in water 2.5 m deep and targeting the USV's two mechanical systems that include rudder and sail-wing actuators powered by electric motors that can radiate noise at low frequencies (<6 kHz). One hydrophone was placed 2 m below the USV before the rudder and then the sail-wing actuators were driven independently for noise testing.

Overtakes

When underway, the USVs moved at lower speeds than the overtaking ships ([Table 1](#)). These speed discrepancies precluded use of a traditional “follow-the-leader” approach, because the powered ships would have had to operate at slower velocities than during a typical acoustic survey, potentially impacting the interpretability of fish responses and associated acoustic data from motorized vessels. Instead, we implemented an “overtake” design whereby survey vessels approached the drone from astern at their standard survey speeds, giving USVs enough lead time to survey at least 2 km before being overtaken. A transect length of 2 km was selected to limit the time between the ship and the USV survey and to allow more than one overtake per night. With a 2 km transect, we expected to observe ~300 single echo detections by the echosounder in Lake Superior, which would give us enough TS measurements to make meaningful contrasts between ships and USVs. Overtaking vessels followed along the path traveled by the drone and surveyed the same water within 120 min (usually within 60 min). Overtaking vessels turned aside when they were within ~50 m of the USV, and avoided passing in front of the drone, which would have entrained large amounts of bubbles into surface waters. Between one and five overtakes were collected during each night, all past astronomical dusk and prior to astronomical dawn.

We analyzed drone–ship overtakes in two ways. First, we compared acoustic measures between the drone and the overtaking vessel surveying the same 2 km transect to test whether ship avoidance can be detected when integrating across a transect. Second, we examined acoustic responses measured by the USV as the vessel approached to evaluate the distance in front of an approaching motorized vessel at which fish responses may be detected.

Acoustic data processing

Vessels and USV data were processed using the same methodology. We first used the best bottom candidate function (Echoview v13.1) to identify the lake bottom down to 100 m, before it was manually reviewed and edited; only 18 of 44 overtakes occurred in water > 100 m deep. To prevent bottom intrusion into the analysis region, we used a 0.5 m back-step added to the identified bottom if it was shallower than 100 m. Most fish were above 100 m at night in Lake Su-

perior ([Stockwell et al. 2006](#)), and therefore fish close to the bottom did not impact our study. All drone and ship transducers were mounted some distance below the surface (0.8–2.3 m, [Table 1](#)). We therefore used a surface exclusion zone of 5.0 m to account for transducer depth and omit nearfield dynamics from the analysis region ([Simmonds and MacLennan 2005](#); [Parker-Stetter et al. 2009](#)). After setting the surface exclusion zone, we manually reviewed and edited the 5.0 m surface exclusion line to ensure that no surface bubbles were incorporated in the analysis. Adjustments to the surface exclusion zone were rarely needed. Crosstalk between transducers occurred when drones and powered vessels were within ~250 m of each other. We identified these crosstalk samples manually and excluded them from the analysis as “no data” regions. The R/V *Kiyi* data also included crosstalk from the ship echosounder. We removed this crosstalk from the R/V *Kiyi* echograms in two ways: (1) an impulse noise filter for volume backscattering strength (S_v) and TS data, and (2) manual removal using “no data” regions. The “Impulse noise removal” variable in Echoview was used for water depths between 30 and 100 m to identify impulse noise using a vertical window size of two samples, a horizontal window size of three pings, and a ping-to-ping change of > 10 dB. No impulse noise removal in water shallower than 30 m was necessary as impulse noise in shallow water did not reach above the applied fish threshold (see below). Data identified as impulse noise were replaced with the mean value of the two adjacent samples. Background noise was removed with the “Background noise removal” filter ([De Robertis and Higginbottom 2007](#)) in Echoview (horizontal extent: 20 pings, vertical extent: 5 samples, maximum noise was adjusted for each platform; [Table 1](#)).

We applied a “Minimum TS threshold” to the S_v data to limit acoustic returns from nonfish targets (e.g., zooplankton and *Mysis diluviana*). This threshold was set to –66 dB, which was 6 dB below the minimum compensated TS of interest (–60 dB). For discussion on using this threshold to minimize contribution from nonfish targets to the S_v data, see [Rudstam et al. \(2009\)](#). The S_v data were divided into one ping by 1 m depth cells and exported to R (v4.3.0; [R Core Team 2023](#)) for further processing. Individual fish targets were identified using Echoview’s “Single target detection for split beams, method 2” with settings suggested in the Great Lakes standard operating procedure (maximum beam compensation was 6 dB, standard variation for minor and major axis angles was 0.6, echo lengths between 0.6 and 1.5 times the pulse duration, and minimum fish target was set to –60 dB; [Parker-Stetter et al. 2009](#)); this algorithm identifies targets within the beam defined by the beam compensation values supplied. Depth and TS of all identified single targets were exported for processing in R.

Analysis of USV acoustic data from overtakes

To explore if and when fish responded to vessels, we fit generalized additive models (GAMs) with five knots to z-scores of nautical area scattering coefficient (NASC, s_A ; $m^2 \cdot nmi^{-2}$; 1 nmi = 1.852 km), average TD (m), and average TS (dB) measured by the drone and calculated at the level of each over-

take (to standardize between overtakes) as a function of the distance to the vessel (see supplementary material for a more detailed description). Distance was evaluated by using 30 s intervals starting when the motorized vessel was 2 km away from the USV and ending when the vessel reached its closest point to the drone (often closer than 40 m). We divided each response into four depth bins (5–20, 20–40, 40–60, and 60–100 m). Intervals where NASC was zero were dropped before z -scores were calculated. For each USV acoustic variable (NASC, TD, and TS) we compared four GAMs fit with different predictors (depth and vessel, vessel only, depth only, and no factor included) using corrected Akaike information criterion (AIC_c) scores; we interpreted a flat line as indicative of a lack of fish response.

Differences between USVs and motorized ship acoustic data

We compared acoustic results between USVs and motorized survey vessels, integrated over 2 km transects, to investigate differences in acoustic fish estimates between the different types of platforms. We analyzed the ship and USV data from each transect as paired data because the same water was surveyed by both platforms within a short period of time (<120 min). As with the USV overtake analysis, we divided the water column into four depth bins starting with the surface bin of 5–20 m followed by up to three depth bins (20–40, 40–60, and 60–100 m) depending on bottom depth. The upper depth bin only included data from 5 to 20 m because of the surface exclusion zone. We excluded depths greater than 100 m from this analysis as fish are rare below 100 m and standard surveys do not collect acoustic data deeper than 100 m in Lake Superior. For the whole transect, we compared the differences in S_A in dB re $1 \text{ m}^2 \cdot \text{nmi}^{-2}$ (calculated as: $\Delta S_A = S_{A, \text{USV}} - S_{A, \text{vessel}}$), the difference in average TD in meters ($\Delta \text{TD} = \text{TD}_{\text{USV}} - \text{TD}_{\text{vessel}}$), the difference in average TS in dB re 1 m^2 ($\Delta \text{TS} = \text{TS}_{\text{USV}} - \text{TS}_{\text{vessel}}$), and the ratio in areal fish density between drone and vessel. Areal fish density was calculated using echo integration, that is by dividing NASC (S_A) values with the average backscattering cross section (σ_{bs}) values in each depth bin and adjusting the unit to fish/ha (i.e., $D = (10^4 \cdot S_A) / (4 \cdot \pi \cdot 1852^2 \cdot \sigma_{\text{bs}})$). We $\log_{10}$ -transformed the fish density ratio (R_D) to normalize: $R_D = \log_{10}(D_{\text{USV}}/D_{\text{vessel}})$. We anticipated that if fish avoided survey vessels, S_A , TS, and TD would be lower and target depth would be deeper in the survey vessel data. We tested each response (ΔS_A , ΔTD , ΔTS , and R_D) with a two-way Analysis of variance (ANOVA) regression with vessel and depth as factors to investigate whether differences in acoustic metrics between motorized ships and the quiet USV varied by vessel or depth layer. Subsequently, if data showed lack of support for vessel or depth layer factors as affecting differences in acoustic metrics between the USV and motorized vessels (ΔS_A , ΔTD , ΔTS , and R_D), we re-analyzed data sets with t tests after pooling data across unsupported factors. We then used the `pwr.t.test` function in the `pwr` package in R (Champely 2017) to evaluate the minimum effect size we could have detected with a power = 0.8 and alpha = 0.05.

Results

Vessel noise

The USV range-corrected radiated noise was more than 50 dB lower than that of the conventional powered vessels in the 100 Hz to 1 kHz frequency range (Table 1). It is likely that the differences are even greater at frequencies less than 100 Hz because the USVs were so quiet that their acoustic signatures were not measurable in that frequency range (Fig. 2). Even when measuring the USV dockside, it was challenging to detect differences from ambient noise (Fig. S1). The noise from the USV was focused on the mid-frequency region (600 Hz to 6 kHz) and had a relatively flat curve compared to the more continuously declining curve in the motorized vessel data (Fig. 2). The R/V *Hack Noyes* and R/V *Kiyi* were the loudest vessels on approach (Table 1) with noise levels 10–13 dB higher than the other two vessels tested. The peaks in received sound pressure levels at low frequencies for the motorized vessel profiles and their harmonics were due to the blade pass frequency (number of blades multiplied by the main shaft rotations per second; Fig. 2). At frequencies greater than 200 Hz, the noise radiating from powered vessels becomes more broadband (Fig. 2) and was likely the result of radiated noise from many mechanical sources on the vessel as well as flow noise from the propeller. Repeated runs for the same vessels typically varied by ~10 dB (7–26 dB, Table 1), likely due to the vessel not passing directly over the hydrophone array, creating angle of arrival differences from run-to-run, and the vessel transiting in the opposite direction for subsequent runs.

Overtake conditions

Forty-four overtakes were conducted, 18 with the R/V *Kiyi*, nine each with the R/V *Bagosendam* and R/V *Hack Noyes*, and four each with the R/V *Blackfin* and R/V *Nomegos* (Table S2). Weather conditions during overtakes were relatively calm, providing good conditions for data collection by both vessels and USVs (Table S2). Wind speeds during overtakes were generally low (mean = $4.5 \text{ m} \cdot \text{s}^{-1}$, range = 0.4 – $17.8 \text{ m} \cdot \text{s}^{-1}$), and the USVs moved at a mean speed of $0.66 \text{ m} \cdot \text{s}^{-1}$ (range = 0.22 – $1.4 \text{ m} \cdot \text{s}^{-1}$). Ships overtook the USVs at speeds ranging from ~2.0 to $2.5 \text{ m} \cdot \text{s}^{-1}$ (Table 1).

Fish behavioral response to approaching vessels

GAMs showed no evidence of sudden changes in acoustic metrics recorded by USVs as ships approached and were more indicative of gradual changes across the transects (Table 2, Table S3). All GAMs fit to TS data were flat lines without slope (Fig. S2), and regardless of model, no smooth term was found to be significant (Table 2). For NASC and TD, the best-supported models were >8 AIC_c units lower than the next best model (Table 2). In the best-supported NASC model (factor: vessel only) the R/V *Nomegos* smooth term was significant (effective degrees of freedom (edf) = 3.1, Ref. df = 3.6, f -value = 4.0, p -value < 0.01; Table S3, Fig. S3), indicating a potential fish response when the R/V *Nomegos* approached. The full model was only best supported when TD was the response (factors: interaction of depth and vessel; Table 2, Table S3).

Fig. 2. The range-corrected sound pressure levels spectra for the five powered vessels in the study averaged over 10 s of a single run when modeled with spherical or cylindrical spreading. In addition, the suggested noise levels proposed by the International Council for the Exploration of the Sea that would limit impact on fisheries acoustic surveys (Mitson 1995) are plotted as a dashed line, and are defined by the following equations:

$$1 \text{ Hz to } 1 \text{ kHz} = 135 - 1.66 \log_{10}(f_{\text{Hz}})$$

$$1 \text{ kHz to } 100 \text{ kHz} = 130 - 22 \log_{10}(f_{\text{kHz}})$$

where f = frequency. Note that both modeling methods (spherical or cylindrical) exceed the Mitson recommendations except in the case of the R/V *Blackfin*, and the actual transmission loss likely lies between these two simple assumptions. Both spherical and cylindrical spreading assumptions are shown as these data were recorded in shallow water (~30 m deep); the true transmission loss is likely more complicated and lies between these two lines. The uncrewed surface vessel (USV) range-corrected sound pressure levels for the rudder and sail-wing actuation conditions are shown in the lower panel. The bold sections of the lines are statistically significant signal-to-noise ratios, and the dotted lines are where the vessel noise cannot be accurately differentiated from the ambient, surface wave-, and wind-generated noise. The received, uncorrected levels for each vessel, along with their respective background measurements, are provided in Fig. S6. The USV noise is only displayed at frequencies greater than 100 Hz because the signal-to-noise ratio below 100 Hz was only background noise.

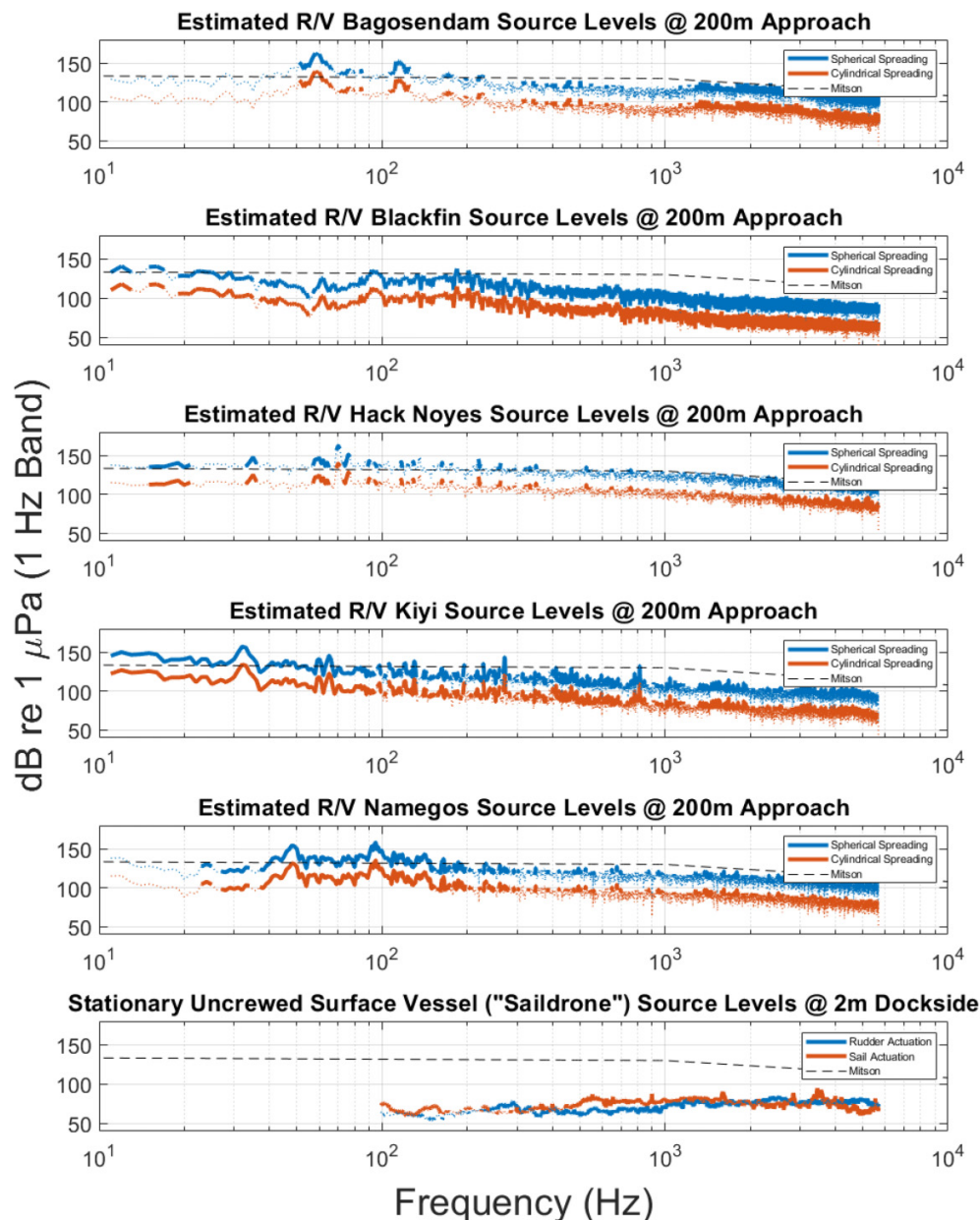


Table 2. General additive models (GAMs) fit to drone acoustic data as motorized ships approached.

Response	Factor(s)	AIC _c	ΔAIC _c	Number of smooth terms	Significant smooth terms
NASC	Vessel	7766.0	0.0	5	1
NASC	None	7774.7	8.7	1	0
NASC	Depth × vessel	7776.7	10.6	20	4
NASC	Depth	7780.2	14.2	4	0
Target depth	Depth × vessel	4379.2	0.0	20	3
Target depth	Vessel	4390.1	10.8	5	2
Target depth	Depth	4398.6	19.4	4	1
Target depth	None	4406.8	27.6	1	0
TS	None	4406.4	0.0	1	0
TS	Depth	4409.3	2.9	4	0
TS	Vessel	4411.1	4.7	5	0
TS	Depth × vessel	4425.1	18.7	20	0

Note: Columns show the acoustic response variable, regression factors included in GAMs, degrees of freedom used by factors in the model, corrected Akaike information criteria (AIC_c), the difference in AIC_c units from the best-supported model (ΔAIC_c), the number of smooth terms estimated by each model, and the number of fitted smooth terms that were supported as statistically significant (i.e., p -value < 0.05). Significant smooth terms indicate a nonflatline signature in drone acoustic data and may be indicative of a response in the acoustic metric as ships approached. TS, target strength; NASC, nautical area scattering coefficient.

In this model, significant smooth terms had complex shapes (edf $\sim$ 3) and included three different vessels and two different depth layers (Table S3, Fig. S4).

Differences in acoustic measures integrated over 2 km survey transects

Overall, overtake data indicated that motorized vessels recorded lower S_A and TS values and slightly lower fish density compared to USVs, but suggest no difference in TD (Fig. 3). Two-way ANOVAs indicated lack of support for systematic differences among survey vessels or depth layers for all metrics (Table 3, Fig. 3). Vessel was identified as a significant factor only in analyzing R_D data; however, post hoc analysis indicated this was restricted to weak support for a single vessel contrast between R/V *Nomegos* and R/V *Kiyi* (p -value = 0.048). Thus, failing to identify systematic variation among vessels or depth layers, we pooled data among these factors and re-analyzed acoustic difference data using t tests to increase our power to detect potential differences between the USV and motorized ship acoustic metrics. Given our sample sizes and the variability we observed, we could have detected a difference of 0.85 dB for ΔS_A , 0.7 m for ΔTD , 1.1 dB for ΔTS , and 0.08 for R_D (i.e., 20% difference in back-transformed fish density; Table S4). T tests on pooled data indicated support for differences for ΔS_A (mean = 2.4 dB, t -value = 7.03, df = 146, p -value < 0.0001) and ΔTS (mean = 1.8 dB, t -value = 4.14, df = 146, p -value < 0.0001) between motorized vessels and USVs, but no differences were detected for ΔTD (mean = -0.1 m, t -value = -0.37, df = 146, p -value = 0.71). The t test on R_D indicated support for nonzero differences (mean = 0.06, t -value = 1.91, df = 146, p -value = 0.058). Although S_A measured by the drones were 2.4 dB higher than measured from the vessels (which equals to 74% higher s_A (NASC) values), this was mainly a result of higher TS values of individual fish (1.8 dB or 51% higher σ_{bs}) rather than a large decrease in

fish density. The difference in fish density was less because fish density depends on the ratio of s_A and σ_{bs} . That said, observed effect sizes for R_D resulted in 14% lower estimates of fish density for motorized vessels than the quiet USVs when calculating fish density using in situ TS.

Discussion

Based on literature from marine surveys (De Robertis et al. 2019; Totland and Johnsen 2022) and the Great Lakes (Grow et al. 2020; Elliott et al. 2021), we hypothesized that the USVs would measure higher nautical area scattering strength (S_A) and higher fish densities than traditional motorized survey ships when comparing the same 2 km transects and that this effect would be larger for louder vessels. We also expected fish would dive to avoid motorized vessels, resulting in deeper targets and a lower TS in the motorized vessel data compared to the USVs. Similarly, as motorized vessels approached the USVs, we expected that fish would show a behavioral response that would lead to a decrease in S_A and fish density and a deeper depth distribution, and this decrease would be related to the noise level of the overtaking survey vessel. Lastly, we expected the strongest responses would be in the upper waters as vessels noises should be louder at these depths. Our results were only partly consistent with these expectations (Fig. 3).

Uncrewed surface vessel S_A values were higher than motorized vessel S_A values (Fig. 3) suggesting a response to the motorized vessels by Lake Superior fishes; however, we could not detect a difference by depth layer suggesting that fish reacted to the ships similarly at depths down to 100 m. This difference averaged 2.4 dB, which represents a 74% higher NASC value obtained by USVs compared to ships. The higher TS values of individual fish measured by the USVs (Fig. 3) resulted in the higher USV S_A values rather than fish avoiding the mo-

Fig. 3. The difference between the uncrewed surface vessel and overtaking motorized vessel integrated over 2 km transects for S_A , target depth, target strength (TS), and the ratio of $\log_{10}$ fish density (R_D) by depth layer for each overtake (points). ΔS_A and ΔTS are in dB units and therefore equivalent to $10 \cdot \log_{10}(NASC_{\text{drone}}/NASC_{\text{vessel}})$ or $10 \cdot \log_{10}(\sigma_{\text{bs, drone}}/\sigma_{\text{bs, vessel}})$, respectively. Target depth is calculated as the difference in mean values (drone – vessel). $\log_{10}$ areal density is calculated as $R_D = \log_{10}(D_{\text{drone}}/D_{\text{vessel}})$. Boxes mark the first–third quartile range, the line is the median, and the whiskers are 1.5 times the adjacent quartile range. NASC, nautical area scattering coefficient.

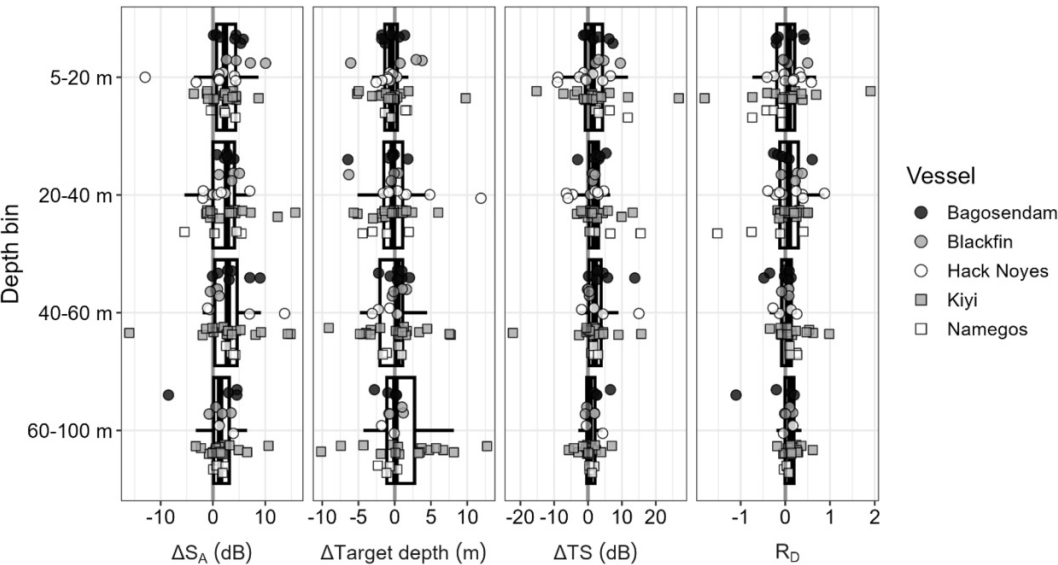


Table 3. Two-way analysis of variance models comparing differences in acoustic data integrated over 2 km transects between motorized vessel and uncrewed surface vessel for ΔS_A , $\Delta \text{target depth}$ (ΔTD), $\Delta \text{target strength}$ (ΔTS), and the $\log_{10}$ ratio of areal fish density (R_D).

Response	Factors	R^2	Significant factor	p -value	Significant contrast
ΔS_A	Depth $\times$ vessel	0.10	None	All > 0.4	None
ΔTD	Depth $\times$ vessel	0.09	None	All > 0.6	None
ΔTS	Depth $\times$ vessel	0.12	None	All > 0.2	None
R_D	Depth $\times$ vessel	0.15	Vessel	Depth $\times$ Vessel: 0.48, depth: 0.95, vessel: 0.053	R/V Namegos–R/V Kiyi ($p = 0.048^*$)

Note: The minimum observed p -value for factor coefficients as a group are reported for regression for which no factors were found to be significant.
^{*} p -value derived from a Tukey’s honest significant test for contrasts within the significant factor (vessel).

torized ships and not being insonified (e.g., by moving horizontally out of the path of the motorized ship). Fish densities were only 14% (0.06 on the $\log_{10}$ scale, R_D) higher with the USVs compared to the ships suggesting that changes in fish orientation are responsible for the increase in S_A observed by the drone. Fish TS is highly dependent on orientation, especially at higher frequencies such as 120 kHz used in the Great Lakes (Simmonds and MacLennan 2005; Rudstam et al. 2012). Because fish density is low enough in Lake Superior to use in situ TS values to calculate fish densities (Parker-Stetter et al. 2009), fish density estimates are less affected by changes in orientation. However, if fish densities were calculated using a standard TS value, as is common practice in marine systems (Simmonds and MacLennan 2005), the difference would have been almost a factor of two, with the drone seeing higher densities than the motorized vessels.

Contrary to our hypothesis that the fish response to motorized vessels would be strongest in the upper water col-

umn (i.e., 5–40 m), we did not detect the expected difference with depth (Tables 2 and 3). However, even 100 m away from the ship is well within the expected range of vessel avoidance in marine systems (Vabø et al. 2002). These results were surprising as a prior study in Lake Superior that compared the R/V Kiyi with a towed sled reported substantially higher avoidance in surface waters than in deeper waters (Grow et al. 2020). The mechanisms driving the differences in upper water column fish responses observed between our finding and the Grow et al. (2020) findings are unclear. Possible factors driving this variability may include differences in the sampling approach between our studies or differences in the fish community present in the two different study years. Our study was conducted in 2022, a year with an unusually high abundance of age-0 cisco, a group of fish that were rare in 2018 when Grow et al. (2020) performed their study. Furthermore, Grow (2023) recently found large differences in fish behavior in 2021 and 2022 using an upward-looking stationary

transducer in Lake Superior and attributed this to the strong cisco year class in 2022. This raises the possibility that fish avoidance behavior varies with age group.

Our third hypothesis that higher ship noise generation would be related to higher avoidance was not supported by our data. The radiated noise of a vessel in the water column alone (Table 1) did not provide a strong predictor of fish avoidance, as vessels were generally indistinguishable during 2 km transects (Table 2). Sound detection is variable and species specific in fishes but most can detect sound at frequencies up to 1 kHz and sometimes as high as 10 kHz, with lower frequencies detected by the lateral line (Mitson 1995; Popper et al. 2022). It is possible that noise levels are not the main trigger of fish avoidance. Silent ships do not always measure higher fish densities than louder vessels (De Robertis et al. 2008). In Lake Erie, DuFour et al. (2021) detected differences among ships, but the higher densities were obtained with the largest and presumably noisiest vessel. However, limiting sound radiated into the water column by scientific vessels can have multiple benefits (Faulkner et al. 2018) such as increasing propulsion efficiency (Merchant 2019). Our work suggests that solutions to minimize potential fish avoidance to acoustic survey vessels, if they occur, may require strategies beyond current sound thresholding recommendations. In this regard, quiet and relatively small USVs hold promise in both reducing fish responses to acoustic survey platforms (Totland and Johnsen 2022) as well as increasing the opportunity to expand survey coverage (Greene et al. 2014; De Robertis et al. 2021).

Our findings of a lack of difference in acoustic metrics among motorized ships may be partially attributable to low statistical power based on the number of transects we were able to complete. Although we conducted a substantial number of overtakes pooled across vessels (44), we were only able to achieve a relatively small sample size of four transects for two of the vessels (R/V *Blackfin* and R/V *Namegos*). Furthermore, overtakes were conducted in different areas of the lake, potentially with different fish communities, which could increase variability in the among-ship comparisons. However, these caveats aside, transect data did not show evidence of systematic differences among ships and we conclude that acoustic data collected with the ships used in Lake Superior are comparable.

Combined, our USV versus ship comparison results suggest that Lake Superior fish react to motorized vessels by changing orientation when the ships are right above them, perhaps to initiate an avoidance reaction, but are still insonified as part of the survey. Even a relatively small change in fish orientation in response to the ships can result in large changes in TS at the higher frequencies used here (120 kHz). Scattering models and empirical measurements both show difference of 10 dB or more associated with changes in orientation depending on the swim bladder orientation relative to the fish main axis (Horne et al. 2000; Frouzova et al. 2005; Henderson et al. 2008; Jech 2011). Such a change in orientation could explain the lower TS of individual fish and therefore the lower S_A values obtained by the ships.

Further, the lack of a response as ships approached the drone supports the notion that the fish behavioral changes

occur as the ship passes directly over the fish and not from fish avoiding ships at longer distances. Lake Superior has substantial commercial and recreational ship traffic and noise levels can be consistently above background outside of winter months (Putland et al. 2022). Fish may therefore be accustomed to ship noise. Interestingly, in a similar study in lakes Michigan and Huron (Evans et al. 2023), we did not find a decline in S_A and TS values in the ship data relative to the drone data. This may be the result of different fish species (in Lakes Michigan and Huron alewife, *Alosa pseudoharengus*, and bloater, *Coregonus hoyi*, predominate).

The USV and motorized vessels operated with different pulse durations because the units do not offer identical pulse durations in their allowable settings; the USV used a pulse duration of 0.256 s and motorized vessels used 0.400 s. Pulse duration does affect resolution of single targets but not area backscattering as pulse duration is accounted for in those calculations. Given the low fish densities and dispersed targets in Lake Superior at night (Fig. S5), this difference in resolution should not affect the results. Prior work with 70 kHz transducers did not find differences when comparing similar pulse durations (0.256 vs. 0.512 s) in lakes (Godlewski et al. 2011), and simultaneous surveys with two different transducers operated with different settings (including pulse duration) produced comparable biomass estimates (Wanzenböck et al. 2003).

The use of autonomous platforms in aquatic applications is growing and development of both surface and underwater vehicles is advancing rapidly (Moustahfid et al. 2020; Schwing 2023). This study builds on prior work examining fish avoidance to motorized survey vessels in the Great Lakes (Evans et al. 2023) and expands the largest study to date in freshwaters exploring this topic. Even though all sound-profiled vessels exceeded the noise levels recommended by ICES for quiet vessels at some frequency (Fig. 2), fish avoidance appeared limited, as was concluded in 2021 for Lakes Michigan and Huron (Evans et al. 2023). These results imply that fish behavioral responses are not simply a function of vessel noise. Given the widespread use of motorized survey ships for fisheries surveys, continued investigation into how fish detect and potentially avoid vessels by using USVs as reference benchmarks could improve acoustic surveys to inform the status and management of fish populations.

Acknowledgements

We thank the numerous partner organizations that contributed to this study and worked closely with us as we developed the study. We are grateful to Vic Santucci, Randy Claramunt, Steve LaPan, Roger Knight, Kurt Newman, John Dettmers, Jeff Tyson, and Russ Strach for their guidance during the conceptualization of the research. We also thank the vessel crews of the R/V *Bagosendam*, R/V *Blackfin*, R/V *Hack Noyes*, R/V *Kiyi*, and R/V *Namegos* who did extensive night work, often with short notice, and without whom this study would not have been possible. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government. The findings and conclusions in this article are those of the author(s) and do not

necessarily represent the views of the U.S. Fish and Wildlife Service. This journal article has been peer reviewed and approved for publication consistent with USGS Fundamental Science Practices (<https://pubs.usgs.gov/circ/1367/>).

Article information

History dates

Received: 13 March 2024

Accepted: 22 July 2024

Accepted manuscript online: 15 August 2024

Version of record online: 15 November 2024

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Data availability

The data used in this study are publicly available through the United States Department of the Interior data repository: <https://doi.org/10.5066/P13SJ9PW>.

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Competing interests

The authors declare there are no competing interests.

Funding information

This research was supported by the U.S. Geological Survey (grant numbers G21AC10348 and G21AC10745).

Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfas-2024-0087>.

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